

Biocide impact on transporter gene expression in the human pathogen *A. baumannii* BAL062

Abstract

Acinetobacter baumannii is a multi-drug resistant pathogen associated with serious nosocomial infections of the immunocompromised. In addition to its antibiotic resistance, this bacteria also displays tolerance to biocides, further increasing its threat in the hospital environment. This study utilised a dataset produced by Li and colleagues to investigate trends of gene expression in response to biocide stress in the clinical isolate *A. baumannii* BAL062. Specifically, expression of the transporter complexes AdeABC, AdeIJK, ZnuABC and Ttg2ABC in response to stress from ten commonly used biocides was investigated. The first hypothesis, that expression profiles of AdeABC and AdeIJK would significantly differ from those of ZnuABC and Ttg2ABC, was supported. AdeABC and AdeIJK expression is more sensitive to biocide stress than ZnuABC and Ttg2ABC, at both an overall protein-complex level as well as between the constituent genes of the complexes. Moreover, the gene expression data indicated that AdeABC and AdeIJK are vital for *A. baumannii* BAL062's tolerance of the biocides benzalkonium and chlorhexidine. Additionally, AdeIJK alone confers tolerance to chloroxylenol and triclosan. The second hypothesis, that the unique biocide-induced expression profiles of genes can be used to speculate on the function of hypothetical proteins *BAL062_00181* and *BAL062_01982*, was also supported. *BAL062_00181* is likely a Ttg2ABC-associated protein and *BAL062_01982* is likely a ZnuABC-associated protein. However, while these functional assessments are not concrete, they can be used to guide future research on the *A. baumannii* pathogen.

Introduction

Acinetobacter baumannii is a human pathogen typically associated with nosocomial infections; particularly in the immunocompromised¹. This bacteria can cause various diseases, such as infections of both skin and soft tissue, UTI's, meningitis, bacteraemia, and pneumonia². Treatment of *A. baumannii* infections often prove difficult due to its resistance to antibiotics². In fact, when considering the cumulative resistance across all strains, *A. baumannii* displays resistance to all

currently used antibiotics². This has led the bacteria to be classified as a “Priority 1 Pathogen” by the World Health Organisation³. In addition to its multi-drug resistance, *A. baumannii* displays tolerance to many commonly used biocides⁴ (disinfectants and antiseptics), further exacerbating its threat to the hospital environment. However, while the antibiotic resistance mechanisms have received much attention, there exists a lack of detailed literature surrounding the bacteria’s response to biocide stress⁴.

A study by Li and colleagues⁴ attempted to fill this knowledge gap by elucidating biocide targets and resistance mechanisms in the clinical isolate, *A. baumannii* BAL062⁵. The study presented a dataset displaying changes in *A. baumannii* gene expression in response to exposure to various biocides. A portion of genes assessed that are of particular interest were those associated with transmembrane transport and efflux. While the targets and mechanisms of many biocides are ill-defined⁴ there is evidence to suggest that some may act intracellularly to cause their biocidal activity⁴. As such, it stands to reason that transporters in the cell membrane may potentially play a role in biocide resistance.

Utilising the dataset referenced, this study investigated the impact of biocide stress on four different transporter complexes: AdeABC, AdeIJK, ZnuABC and Ttg2ABC. These transporters are all tripartite pumps, meaning each are composed of three *A*, *B* and *C*. genes However, the structure of these pumps vary. Transporters AdeABC and AdeIJK are both transmembrane resistance-nodulation-cell division (RND) pumps associated with multi-drug resistance (MDR)². In contrast, ZnuABC and Ttg2ABC are both inner-membrane ATP-binding cassette (ABC) transporters, associated with Zn²⁺ transport⁶ and MDR and toluene tolerance⁷, respectively. It was hypothesised that these pumps will have unique expression profiles resulting from biocide stress. More specifically, AdeABC and AdeIJK were predicted to have similar expression profiles, as both are RND transporters, and that these profiles will be distinct from the ABC transporters, ZnuABC and Ttg2ABC.

The dataset used for this investigation also assessed numerous hypothetical (undescribed) proteins. Initial exploratory analysis suggested that some of these hypothetical proteins may be associated with some of the transporter families. To investigate this association, a second hypotheses was tested: That tripartite transporters, and their associated genes, have unique biocide expression profiles. Such profiles can be used to speculate the function of two hypothetical proteins, *BAL062_00181* and *BAL062_01982*.

Methods

Dataset

The dataset produced by Li and colleagues⁴ is the result of a transcriptomics investigation utilising the TraDIS protocol^{4,8}. This dataset contains differences in expression of each protein-coding gene between ten biocide treatments (AgNO₃: silver nitrate, BZK: benzalkonium, CTAB: cetyltrimethylammonium bromide, CHL: chlorhexidine, TRC: triclosan, CRL: chloroxylonol, PVPi: polyvidone iodine, HClO: sodium hypochlorite, GLU: glutaraldehyde, EtOH: ethanol) and a non-treatment control group. Such changes in gene expression (or more specifically, changes in the number of mRNA transcripts detected for each gene in TraDIS sequencing) are expressed as a log fold-changes (logFC) in expression, with the non-treatment group acting as the reference.

Due to the high-fidelity of TraDIS sequencing, this dataset contains genes from the entire *A. baumannii* BAL062 genome; which was impractical to work with. However, the authors had subset the data into smaller sets for genes with related functions. This investigation utilised the *Transporters* subset, which contained 30 genes associated with transporter proteins, their regulators and other associated proteins.

The dataset also contains locus tags of each gene in *A. baumannii* BAL062, the locus tag for the corresponding gene in the model strain, *A. baumannii* AB5075⁹, and the names and functions of each gene (where known) as described by the European Molecular Biology Laboratory Nucleotide Sequence Database (EMBL-Bank)¹⁰ (following typical TraDIS protocol⁸). Significance values (p-values and q-values) for each logFC measurement are also listed. For the purposes of this investigation, this dataset was filtered to solely retain the gene names (independent variable), biocide names (depend variable, 10 levels, fixed effect) and associated logFC values. A grouping factor of gene family was manually added to group each trio of *A*, *B* and *C* genes into their respective families (for example, genes *adeI*, *adeJ* and *adeK* were grouped into the AdeIJK family). logFC values for each family were calculated as an average of the three constituent genes. While all 30 *Transporters* genes were included in the initial exploratory analysis, subsequent analysis only considered the 12 transporter family genes, unless otherwise stated.

All following data analysis was performed using the software package R version 4.4.3 (R Development Core Team 2025)

ANOVA Assumptions

For each ANOVA conducted, the ANOVA assumptions were first tested to assess the validity of the resultant statistics. Homogeneity of variance was assessed by a Levene's Test, which tests the null hypothesis that the variance is equal across all groups. This was further tested by plotting residuals against fitted values and assessing the fitted line. The assumption of normal data was tested by assessing the linearity of a Normal Q-Q plot. The third assumption, the independence of observations, is assured by the experimental design. The expression of each gene transporter family is regulated by independent and unique systems, ensuring that the expression of one family is not dependent on the expression of another.

Due to issues with meeting the ANOVA assumptions, normalisation of the logFC data was attempted. The data was treated as a Student's t distribution (with 9 degrees of freedom) and normalised using R's *qnorm()* function. However, due to resultant issues in logFC interpretability, this normalisation was ultimately abandoned.

Hypothesis 1: Differences between transporter families

A Principal Components Analysis was conducted and used to construct a scree plot. This was accompanied by a Horn's Parallel Analysis to seek further confirmation on unclear results. Based on these plots, factor analyses utilising 5 (as indicated by the Horn's Parallel Analysis) and 4 (as indicated by the scree plot) factors were performed and the factor analysis scores were plotted. These plots were used as an initial exploratory analysis to determine if genes of the same transporter family displayed correlation.

To investigate differences in biocide-induced logFC between gene families, a One-way ANOVA was performed. This ANOVA was followed by a Tukey's test of Honestly Significant Differences (HSD) to characterise the variation between groups. The relationships highlighted by this analysis were visually supported by the construction of bar plots displaying the biocide-induced expression profiles for each gene family.

These bar plots were also used for identification of "essential genes" for biocide tolerance. Such assessment of gene function is a product of the TraDIS methodology. Briefly, DNA sequencing of cDNA fragments (produced via Reverse Transcription-PCR of the mRNA transcripts used to detect expressed genes) is directed by Tn5 transposon insertions throughout the genome¹¹. However, these transposons can disrupt proper gene function. As a result, if disruption of a particular gene results in the death of cells under a given treatment (in this instance, biocide stress), the disrupted gene is removed from the detection pool. This manifests as a dramatic decrease in expression when compared to the non-treatment group, resulting in a large and negative logFC¹¹. Following typical

TraDIS convention, gene families with a $\logFC \leq -2$ were considered “essential” for surviving the stress of a given biocide¹¹.

Hypothesis 2: Utilising biocide profiles to characterise hypothetical proteins

The before-mentioned 4-factor analysis suggested that hypothetical protein *BAL062_00181* was associated with ZnuABC, and that *BAL062_01982* was associated with Ttg2ABC. The plot grouped the known transporter-associated genes *zur* (a ZnuABC transcription repressor⁴) and *mlcA* (a substrate-binding protein separate from, but interacting with, Ttg2ABC⁴) with the *znuABC* and *ttg2ABC* genes, respectively. To further investigate this relationship, logFC data for *BAL062_00181*, *BAL062_01982*, *zur* and *mlcA* were included in another One-way ANOVA that investigated the interaction between gene groups (a stand-in for transporter families) and biocide-induced expression profiles. These interactions were further characterised by a Tukey’s HSD. Finally, the logFC values for each hypothetical protein and the known associated gene were plotted alongside each gene family for a visual assessment of biocide profile similarities.

Results

Hypothesis 1: Differences between transporter families

The Principal Components Analysis of the *Transporters* dataset was used to construct a scree plot (S1). Visual assessment of the plot indicated four factors to be sufficient. However, the Horn’s Parallel Analysis plot indicated that five factors, rather than four, is preferable (S2). Due to the conflicting results, factor analyses considering both 4 and 5 factors were conducted, and the factor scores were plotted. The 4-factor analysis rejected the null hypothesis that four factors are sufficient to explain the variation in the data ($\chi^2_{11} = 21.21$, $p = 0.0313$). However, the 5-factor analysis did not reject the null hypothesis that five factors are sufficient ($\chi^2_5 = 7.12$, $p = 0.212$). While such results indicate the 5-factor model to be favourable, such a notion is brought into question when considering the plots of the factor analysis scores. The 5-factor plot (Fig 1.) displayed a distinct structure that separated the two iron transporter genes, *feoB* and *BAL062_3669*⁴, away from the rest of the genes. While this distinction is interesting, the mechanisms driving this difference are beyond the scope of this study. Moreover, while the constituent genes of the transporter families do appear somewhat clustered, their crowding alongside the other genes makes such grouping difficult to interpret. In contrast, the 4-factor plot (Fig 2.) clusters the genes in a more easily interpretable manner. While

FIG 1. Plot of factor analysis scores (factors = 5) for all genes of the *Transporters* dataset. Genes from the same transporter family are coloured accordingly (AdeABC: maroon, AdeIJK: aquamarine, ZnuABC: sea green, Ttg2ABC: coral). The genes of the same family appear within the same vague groupings. However, the main structure displayed by this 5-factor analysis is the distinction of *feoB* and *BAL062_03669* (a hypothetical *feoA* domain protein⁴) from the rest of the transporter genes. While this represents that iron transporters have unique biocide-induced logFC profiles, the reasons behind such differences are beyond the scope of this study.

It should also be noted that while the Horn's Parallel Analysis report indicated 5 factors to be sufficient, the fifth factor had an adjusted eigenvalue of 0.143 (3sf) whereas the fourth factor had an adjusted eigenvalue of 0.263 (3sf). As the cut-off for consideration as a significant factor is an eigenvalue > 0 , this fifth factor was only just barely considered significant. It should also be noted that the 4-factor analysis found 72.5% of cumulative variation explained, whereas the 5-factor analysis found 79.7% of cumulative variation explained; an increase of only 7.2%. Such descriptive statistics indicate this fifth factor, while statistically significant, could reasonably be considered negligible. This, coupled with the factor analysis score plots, led to the 4-factor model being accepted for this study. With this in mind, it can be stated that the factor score plot of the data acted as a first hint towards transporter-specific trends in the biocide-induced logFC profiles.

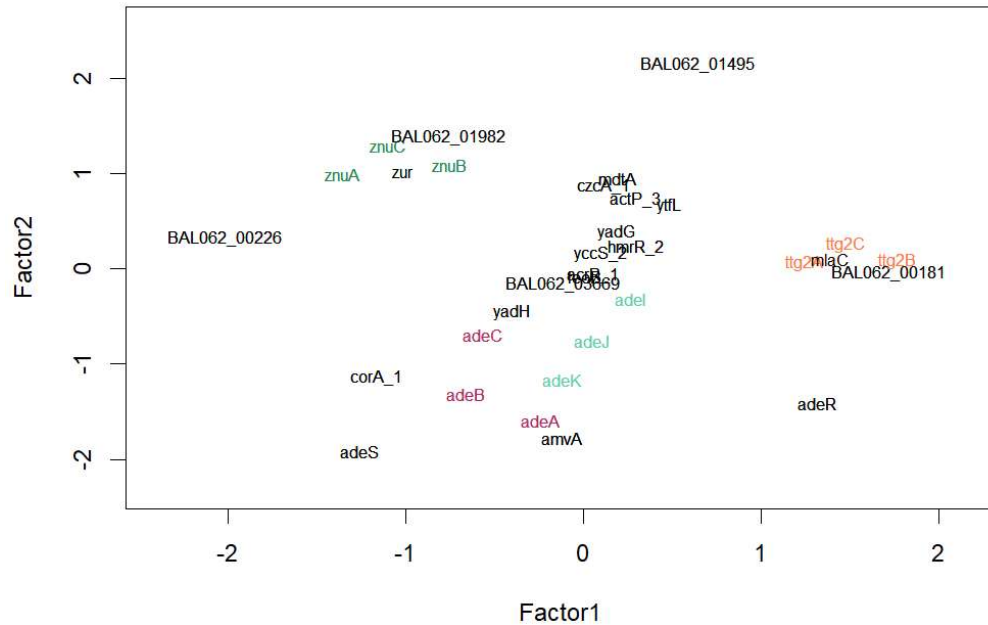


FIG 2. Plot of factor analysis scores (factors = 4) for all genes of the *Transporters* dataset. Genes from the same transporter family are coloured accordingly (AdeABC: maroon, AdeIJK: aquamarine, ZnuABC: sea green, Ttg2ABC: coral). The genes of the same family appear within the same groupings, thus reflecting their related gene functions and hinting towards similar biocide-induced logFC profiles.

this curve (S3). Moreover, the Levene's Test failed to reject the null hypothesis ($F_{4,145} = 2.56$, $p = 0.04107$). Together, this indicated inhomogeneous variance across the gene families. Normality of data was also rejected by the Normal Q-Q plot (S4). Linearity was observed within the first quantile but was lost past this point.

In an attempt to normalise the data, the logFC values for each gene family were treated as a Student's t distribution and subsequently normalised using the *qnorm()* function. While this resulted in more appropriate Residuals vs Fitted (S5) and Normal Q-Q plots (S6), the nature of the logFC scores was altered in such a way that led to misinterpretation of results. It should also be noted that such failures of the strict ANOVA assumptions should be expected from this dataset. As the mean logFC for each gene family is calculated from only three "samples" (the *A*, *B* and *C* genes) the data, by definition, won't be normal. Furthermore, the lack of homogeneous variances can be attributed to the fact that each individual gene of each family had unique logFC values in response to the biocides. However, as meaningful qualitative information can be obtained from the analysis, the ANOVA was still conducted; albeit with the recognition that descriptive statistics will carry an element of doubt and should be interpreted with great care.

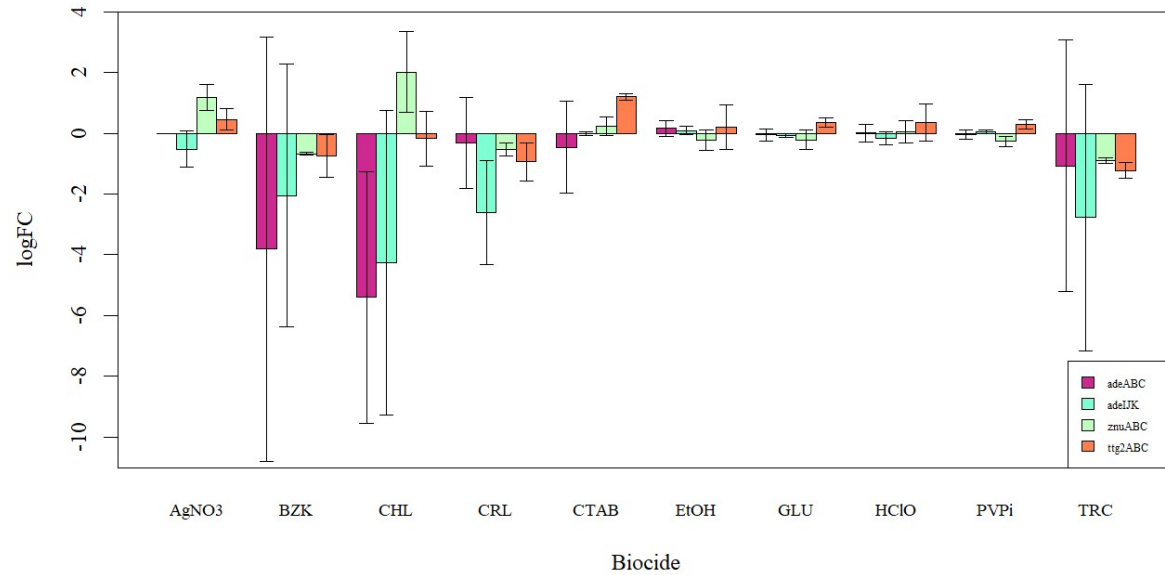


FIG 3. Mean ($\pm 95\%$ CI) logFC measurements ($n = 3$) of transporter families AdeABC, AdeIJK, ZnuABC and Ttg2ABC in response to 10 biocide treatments. Note that the logFC value for each biocide is the average logFC of the three constituent *A*, *B* and *C* genes in each family. The variation between these genes manifests as large 95%CI. The overall pattern of logFC values across the 10 biocides are referred to as the “logFC profiles” or “biocide-induced expression profiles” throughout the text.

The null hypothesis, that the biocide-induced logFC profiles do not vary between the four transporter families, was rejected ($F_{3,116} = 6.5688$, $p = 0.0003854$). Tukey’s HSD found the logFC profile of Ttg2ABC to be 1.08 points higher on average than that of AdeABC ($p = 0.027$, 95% CI [0.0864, 2.07]) and 1.21 points higher than AdeIJK ($p = 0.010$, 95% CI [0.218, 2.21]). ZnuABC was 1.17 points higher than AdeABC ($p = 0.01$, 95% CI [0.177, 2.17]), and 1.30 points higher than AdeIJK ($p = 0.0048$, 95% CI [0.309, 2.30]). However, no significant difference was observed between adeABC and adeIJK ($p = 0.99$, 95% CI [-1.12, 0.862]), nor between ZnuABC and Ttg2ABC ($p = 0.9952$, 95% CI [-0.903, 1.08]). Therefore, the null hypothesis that the profiles of AdeABC and AdeIJK would not be distinct from those of ZnuABC and Ttg2ABC, was rejected.

Such similarities and differences between the transporter families are visualised in **Fig 3**. The distinction between the RND transporters (AdeABC and AdeIJK,) and the ABC transporters (ZnuABC and Ttg2ABC) is attributed to both the directionality of logFC (positive or negative change in gene expression) as well as the magnitude of these logFC scores.

The negative logFC values for the RND transporters extended beyond simple decreases in expression and into the realm of indicators of essentiality. Both AdeABC and AdeIJK displayed $\logFC \leq -2$ for

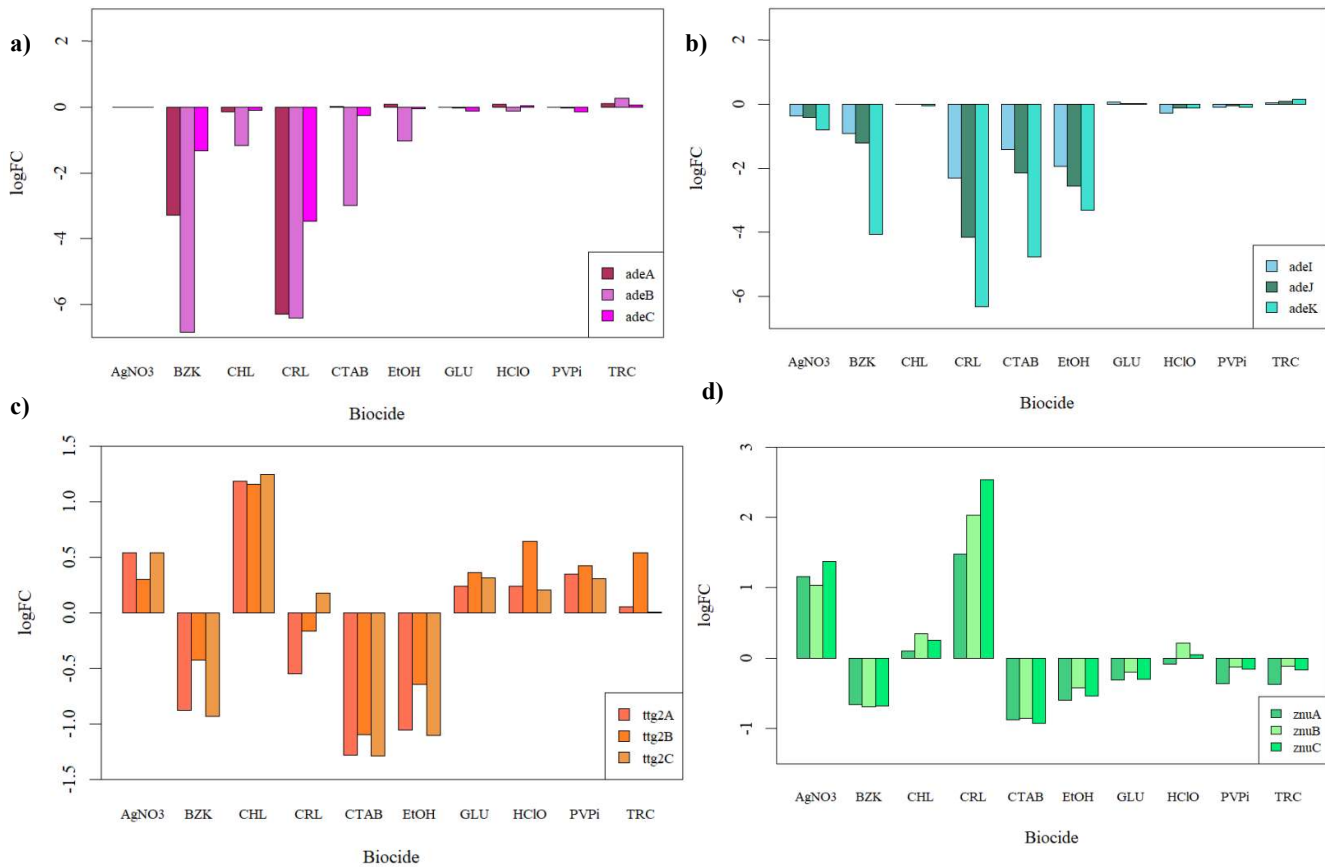


FIG 4. Biocide-induced logFC expression profiles for the *A*, *B* and *C* genes of each transporter family. **a)** AdeABC, **b)** AdeIJK, **c)** Ttg2ABC and **d)** ZnuABC

both BZK and CHL. Additionally, AdeIJK also displayed $\log\text{FC} \leq -2$ for the biocides CRL and TRC.

Another point of difference between the transporter families is the variation of logFC within each family, as suggested by the 95% confidence intervals in **Fig 3**. Intervals for the RND transporters encompassed a wide range, whereas intervals for the ABC transporters were more restricted. This indicated a greater variation in logFC of constituent genes within RND families than those of the ABC families. Such variations are displayed in **Fig 4**. While variation exists across all gene families, it is visually apparent that such variation is more explicit in AdeABC and AdeIJK than in ZnuABC and Ttg2ABC.

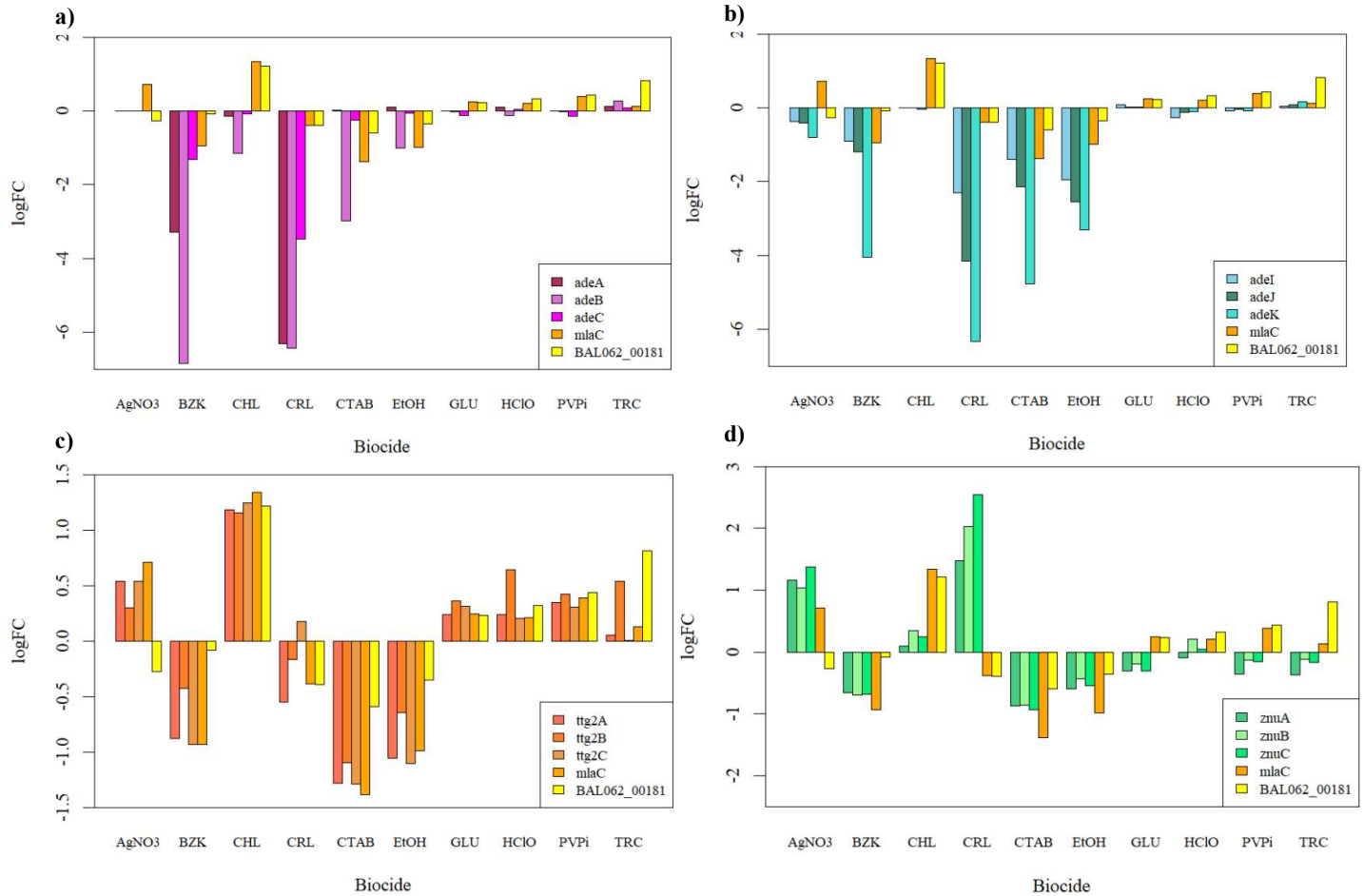


FIG 5. Comparison of biocide-induced logFC profiles of the hypothetical protein *BAL062_00181* and the Ttg2ABC-associated protein, *mlaC*, against each transporter family: **a)** AdeABC, **b)** AdeIJK, **c)** Ttg2ABC and **d)** ZnuABC.

Hypothesis 2: Utilising biocide profiles to characterise hypothetical proteins

A linear model containing the before-mentioned transporter families, with the addition of *zur*, *mlaC*, *BAL062_00181* and *BAL062_01982*, was constructed. As before, trios of related genes were grouped into their respective gene families. However, rather than prematurely associating them with the transporter suggested by the factor score plot (**Fig 2**), the new genes were each treated as their own “family”, or in this instance, “group”. This linear model was tested for the first two ANOVA assumptions. Once again, Residual vs Fit and Normal Q-Q plots displayed that the model failed to meet the ANOVA assumptions (**S7, S8**). However, the Levene’s Test failed to reject the null hypothesis ($F_{7,152} = 1.38$, $p = 0.2162$). Thus, the following descriptive statistics can be regarded with some (while minimal) validity.

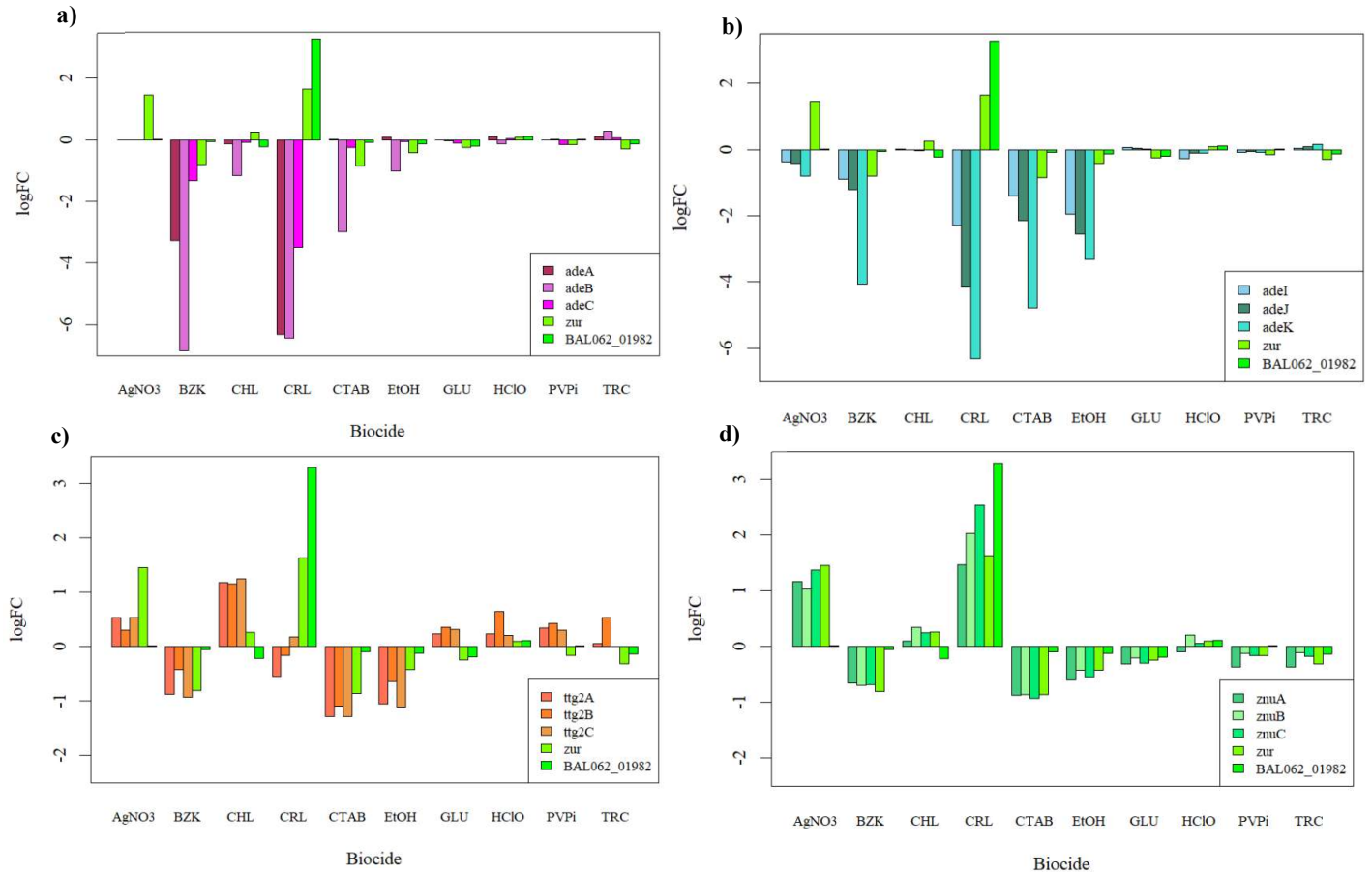


FIG 6. Comparison of biocide-induced logFC profiles of the hypothetical protein *BAL062_01982* and the ZnuABC-associated protein, *zur*, against each transporter family: **a)** AdeABC, **b)** AdeIJK, **c)** Ttg2ABC and **d)** ZnuABC.

A One-way ANOVA once again rejected the null hypothesis that the logFC profiles of the gene groups do not significantly differ from one another ($F_{7,152} = 4.43$, $p = 0.0001649$). These results serve to confirm that adding these additional genes into the model did not disrupt the previously established distinction between gene expression profiles of different transporters.

A Tukey's HSD once again identified significant differences between the transporter families. However, it did not identify any significant differences in expression profiles between the known associate genes (*zur* and *miaC*), nor the hypothetical proteins (*BAL062_00181* and *BAL062_01982*), and the pre-established gene families. While this helps to confirm that these genes do indeed share expression profile similarities to the gene families indicated by the factor analysis plot (Fig 2.), it does not help to differentiate them from the other gene families not suggested by this plot. However, such comparisons are made possible by assessment of the logFC bar plots.

Fig 5. displays that for all biocides but AgNO_3 , *BAL062_00181* and *miaC* have very similar logFC profiles. For gene families AdeABC (Fig 5a), AdeIJK (Fig 5b) and ZnuABC (Fig 5d), these logFC

measurements are visually distinct from the surrounding values. However, these measurements do follow the general logFC trend across biocides for Ttg2ABC (**Fig 5c**).

A similar phenomenon appears in **Fig 6**, for *BAL062_01982* and *zur*. Both genes are similar in the magnitude and directionality of their logFC measurements across most biocides (with the exception of AgNO₃ and CHL). These genes are visually distinct from the logFC trends of families AdeABC (**Fig 6a**), AdeIJK (**Fig 6b**) and Ttg2ABC (**Fig 6c**), but are highly similar to those of ZnuABC (**Fig 6d**).

Such visual plots do not lend themselves to the descriptive statistics of a null hypothesis test. However, based on the before-mentioned observations, it is reasonable to reject the null hypothesis that the logFC biocide profiles cannot be used to aid in confirming the identity of putative transporter-associated proteins.

Discussion

Hypothesis 1: Differences between transporter families

The first hypothesis, that the transporters will have unique biocide-induced expression profiles, was overwhelmingly supported. The hypothesis that the RND transporters, AdeABC and AdeIJK, would have similar expression profiles, and that these profiles would be distinct from those of the ABC transporters, ZnuABC and Ttg2ABC, was also supported. The biocide-induced logFC responses of AdeABC and AdeIJK are more similar to another than to those of ZnuABC and Ttg2ABC. Notably, the RND transporters had more dramatic decreases in expression, whereas the ABC transporters were either relatively unaffected, or upregulated, by biocide stress.

The variation in logFC profiles within each gene family was another point of difference between the RND and ABC transporters. Whereas variation within each ABC transporter was relatively reserved and similar across biocides between ZnuABC and Ttg2ABC, such was not the case for the RND transporters. AdeABC and AdeIJK both displayed great variations in logFC between their constituent genes for the biocides BZK, CRL, CTAB and EtOH, as well as CHL for AdeABC. Moreover, while both RND families displayed an outlier gene with a visually distinct logFC score to its two associates, the particular gene that acted as the outlier differed between the two transporters. In the case of AdeABC, the *B* gene, *adeB*, was the more consistent outlier, whereas the *C* gene, *adeK*, was the outlier of AdeIJK. Such variation between the two RND transporters is particularly interesting when considering the relative positions and functions of each constituent gene. The *B* gene of an

RND pump (*adeB* or *adeI*) is located in the inner membrane and periplasmic space of double membrane cell wall of a the gram-negative bacteria, such as *A. baumannii*¹². In contrast, the *C* gene of an RND pump (*adeC* or *adeK*) is located at the outer membrane of the double membrane structure¹². The fact that such logFC outlier genes differ between the two RND transporters indicates fundamental differences in the functionality and mechanisms of the two transporters. While their overall responses to biocide stress may be similar, the driving forces responsible for such similarities are quite distinct.

It should also be noted that in some instances, the dramatic decreases in logFC of the RND transporters indicated their essentiality for tolerance of certain biocides. Both AdeABC and AdeIJK were considered essential for BZK and CHL tolerance, and AdeIJK alone was also considered essential for tolerance of CRL and TRC. These transporters, previously associated with antibiotic resistance², also appear to confer resistance against biocides. Such findings indicate the versatility of substrates with which such transporters can interact. Further investigations that compare the chemical interactions and binding mechanisms of these biocides with other known substrates of the RND transporters, such as tigecycline¹³ or trimethoprim¹⁴, may provide further insight into the resistance mechanisms of this dangerous MDR pathogen.

Hypothesis 2: Utilising biocide profiles to characterise hypothetical proteins

The second hypothesis, that the unique biocide-induced gene expression profiles of the genes and gene families could be used to speculate the function of two hypothetical proteins, *BAL062_00181* and *BAL062_01982*, was also supported. While descriptive statistics such as Tukey's HSD failed to highlight such functions of these hypothetical proteins, such relationships were interpretable through visual plots. The 4-factor analysis acted as a strong initial line of evidence that tied *BAL062_00181* to the Ttg2ABC family, and *BAL062_01982* to the ZnuABC family. This was further supported by the logFC bar plots in which the profiles of the known associated genes and the hypothetical proteins were compared against the profiles of each transporter family.

The expression profile of *BAL062_00181* displayed a high degree of similarity to the known Ttg2ABC-associated gene, *mlaC*, as well as the to the overall expression profile of Ttg2ABC. This, coupled with the clustering patterns in the 4-factor plot, made it reasonable to conclude that the hypothetical protein *BAL062_00181* is a Ttg2ABC-associated protein. Similarly, the profile of *BAL062_01982* displayed a degree of high similarity to the known ZnuABC-associated gene, *zur*, as well as the to the overall expression profile of ZnuABC. This, along with the grouping of the 4-factor

plot, made it reasonable to conclude that the hypothetical protein *BAL062_01982* is a ZnuABC-associated protein.

However, it is important to recognise that attributing functions to hypothetical proteins in such a manner is not without its limitations. The 4-factor plot, while helpful in guiding the direction of this study, is not an entirely reliable representation of gene associations. For example, the genes *adeR* and *adeS* are both known associated genes of the AdeABC family, acting as transcription regulators¹³ in a manner almost comparable to that of *zur* on the ZnuABC complex⁴. However, these genes were not closely clustered to the *adeABC* genes in the way that *zur* is to the *znuABC* genes in the factor scores plot. While this may cast doubt on the reliability of the factor plot and the subsequent functional identifications of *BAL062_00181* and *BAL062_01982*, it would be improper to say that it disproves these results entirely. After all, such differences between *adeRS* and *zur* in their association with their parent family may simply result from differences in their regulatory mechanisms, or other genetic factors outside the scope of this study. Instead, the assignment of *BAL062_00181* and *BAL062_01982* as associates of Ttg2ABC and ZnuABC should be used to guide further wet-lab research of their function. For example, the creation of a *BAL062_00181* knockout mutant and subsequently assessing Ttg2ABC activity is one potential approach from which *BAL062_00181* function could be inferred. This study's bioinformatic approach to assigning gene function does not attempt to replace lab-based investigations but instead serves to display the ways in which such *in silico* investigations of a single dataset can help inform the directions of future investigations.

Conclusion

This study investigated the nature of biocide-induced gene expression of transporter complexes in the MDR pathogen, *A. baumannii*. The first hypothesis, that expression profiles would differ between RND and ABC transporters, was supported. AdeABC and AdeIJK expression is more sensitive to biocide stress than ZnuABC and Ttg2ABC, at both an overall complex level as well as between the constituent genes of the complexes. Moreover, the gene expression data indicated that AdeABC and AdeIJK are vital for *A.baumannii* BAL062's tolerance of the biocides benzalkonium and chlorhexidine. Additionally, AdeIJK alone confers tolerance to chloroxylenol and triclosan. The second hypothesis, that the unique biocide-induced expression profiles of genes can be used to speculate on the function of hypothetical proteins *BAL062_00181* and *BAL062_01982* was also supported. *BAL062_00181* is likely a Ttg2ABC-associated protein and *BAL062_01982* is likely a

ZnuABC-associated protein. However, while these functional assessments are not concrete, they can be used to guide future research on the *A. baumannii* pathogen.

Acknowledgements

I would like to acknowledge the aid of Professor John Alroy. His help with the initial exploratory analysis, as well as his feedback on handling the ANOVA assumptions, was crucial to the success of this investigation.

Data Availability

All datasets and R scripts used to conduct analysis are available here:

https://github.com/isaak11salami/IsaakSalami_NSCI7915_biocides_genes

References

1. Peleg, A. Y., Seifert, H. & Paterson, D. L. *Acinetobacter baumannii*: emergence of a successful pathogen. *Clin. Microbiol. Rev.* **21**, 538–582 (2008).
2. Morris, F. C., Dexter, C., Kostoulas, X., Uddin, M. I. & Peleg, A. Y. The Mechanisms of Disease Caused by *Acinetobacter baumannii*. *Front. Microbiol.* **10**, 1601 (2019).
3. WHO publishes list of bacteria for which new antibiotics are urgently needed.
<https://www.who.int/news/item/27-02-2017-who-publishes-list-of-bacteria-for-which-new-antibiotics-are-urgently-needed>.
4. Li, L. *et al.* Systematic analyses identify modes of action of ten clinically relevant biocides and antibiotic antagonism in *Acinetobacter baumannii*. *Nat. Microbiol.* **8**, 1995–2005 (2023).
5. Shashkov, A. S. *et al.* Characterization of the carbapenem-resistant *Acinetobacter baumannii* clinical reference isolate BAL062 (CC2:KL58:OCL1): resistance properties and capsular polysaccharide structure. *mSystems* **9**, e00941-24 (2024).

6. Hesse, L. E., Lonergan, Z. R., Beavers, W. N. & Skaar, E. P. The *Acinetobacter baumannii* Znu System Overcomes Host-Imposed Nutrient Zinc Limitation. *Infect. Immun.* **87**, e00746-19 (2019).
7. García, V. *et al.* Functional analysis of new transporters involved in stress tolerance in *Pseudomonas putida* DOT-T1E. *Environ. Microbiol. Rep.* **2**, 389–395 (2010).
8. Barquist, L. *et al.* The TraDIS toolkit: sequencing and analysis for dense transposon mutant libraries. *Bioinformatics* **32**, 1109–1111 (2016).
9. Jacobs, A. C. *et al.* AB5075, a Highly Virulent Isolate of *Acinetobacter baumannii*, as a Model Strain for the Evaluation of Pathogenesis and Antimicrobial Treatments. *mBio* **5**, e01076-01014 (2014).
10. Stoesser, G. *et al.* The EMBL Nucleotide Sequence Database. *Nucleic Acids Res.* **30**, 21–26 (2002).
11. Langridge, G. C. *et al.* Simultaneous assay of every *Salmonella Typhi* gene using one million transposon mutants. *Genome Res.* **19**, 2308–2316 (2009).
12. Du, D. *et al.* Multidrug efflux pumps: structure, function and regulation. *Nat. Rev. Microbiol.* **16**, 523–539 (2018).
13. Xu, C. F., Bilya, S. R. & Xu, W. adeABC efflux gene in *Acinetobacter baumannii*. *New Microbes New Infect.* **30**, 100549 (2019).
14. Damier-Piolle, L., Magnet, S., Brémont, S., Lambert, T. & Courvalin, P. AdeIJK, a resistance-nodulation-cell division pump effluxing multiple antibiotics in *Acinetobacter baumannii*. *Antimicrob. Agents Chemother.* **52**, 557–562 (2008).

Supplementary Information

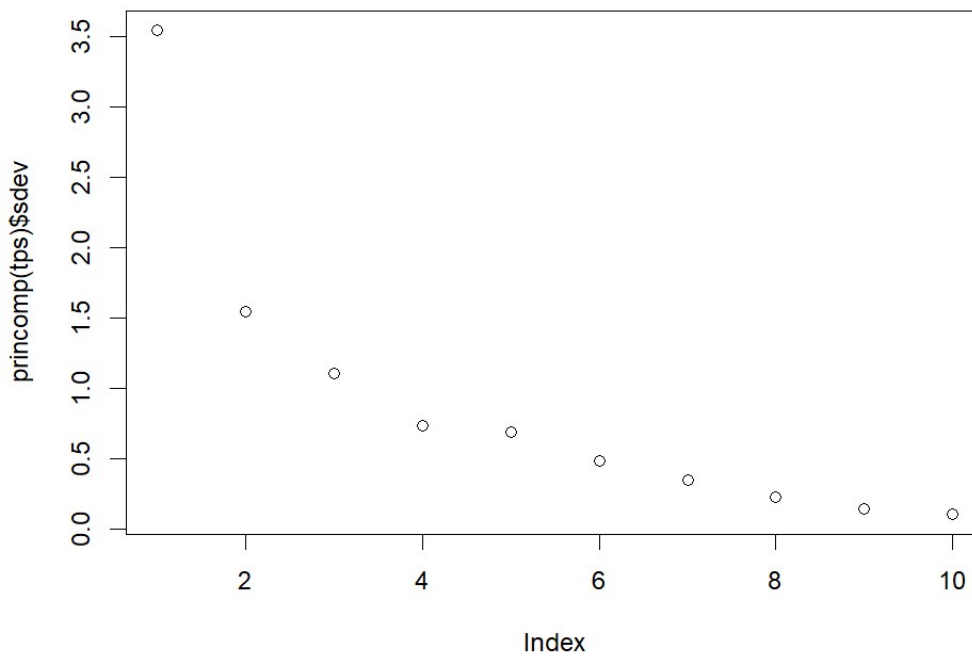


Figure S1. Scree plot of the *Transporters* dataset following Principal Components Analysis. As the curve flattens between the fourth and fifth datapoint, this indicates that a factor analysis should consider 4 factors.

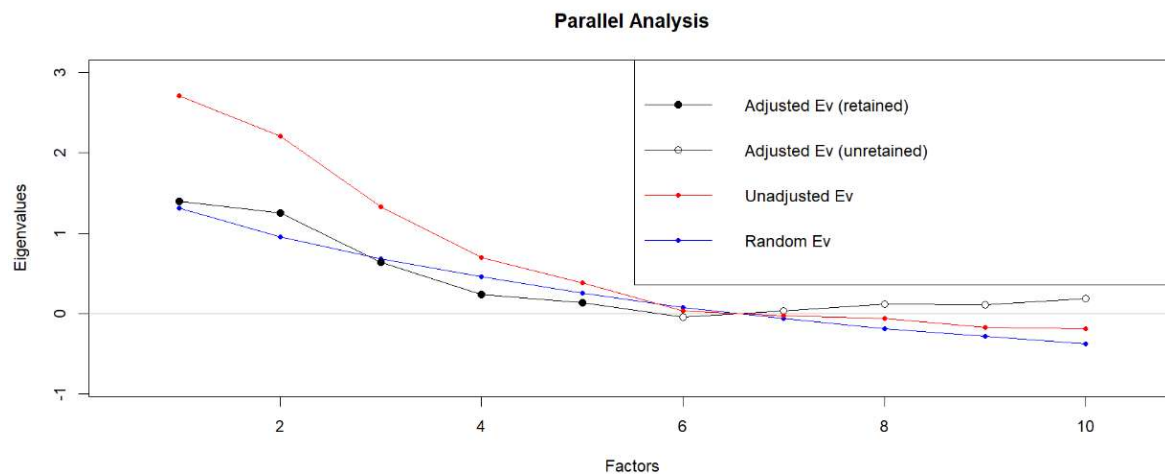


Figure S2. Horn's Parallel Analysis plot of *Transporters* dataset. Of the 10 factors assessed, only the first 5 have eigenvalues greater than 0. Therefore, this indicates that a subsequent factor analysis should consider 5 factors.

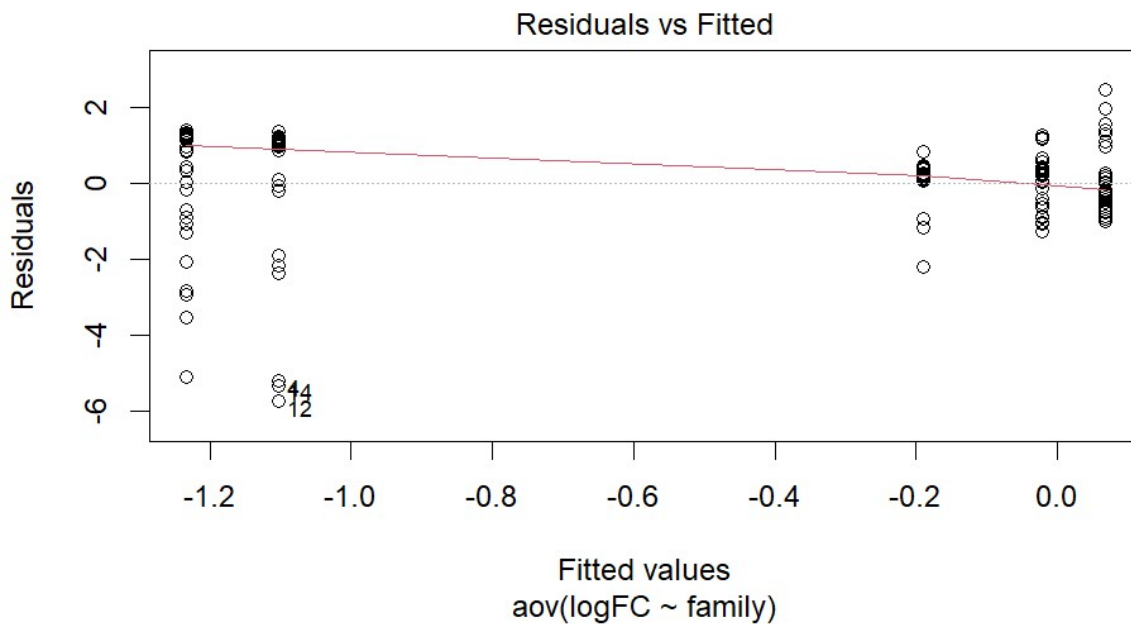


Figure S3. Residuals vs Fitted plot for the ANOVA assumptions testing of the “logFC against gene family” linear model. The negative slope and unbalanced spread of residuals indicates inhomogeneous variance, and thus a failure of ANOVA assumptions.

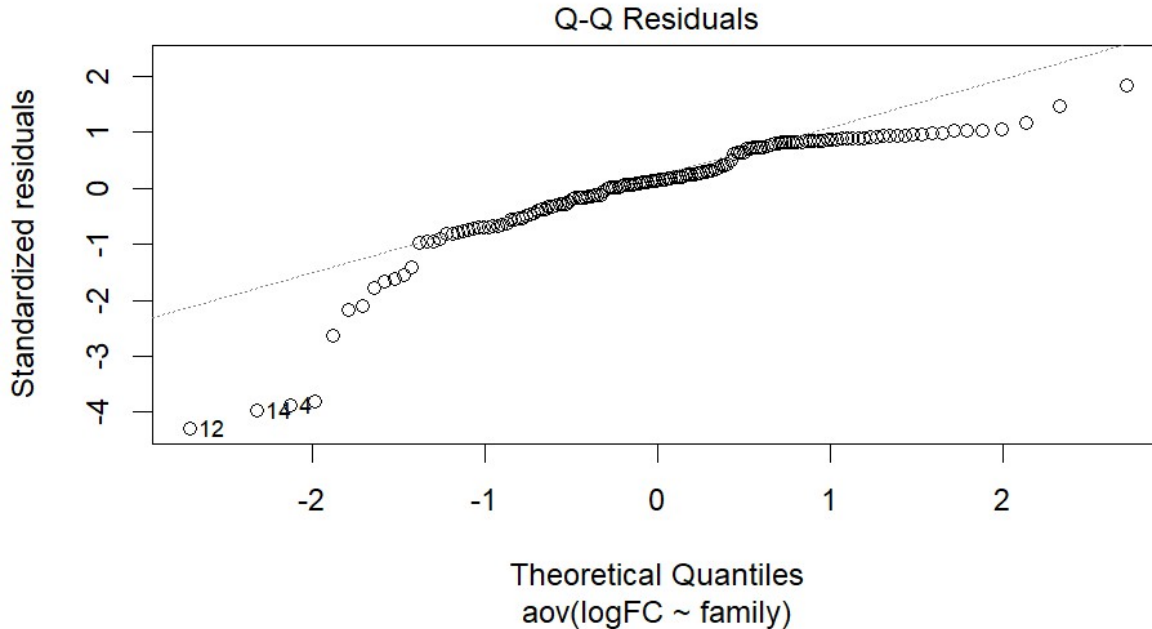


Figure S4. Q-Q plot for the ANOVA assumptions testing of the “logFC against gene family” linear model. The lack of linearity indicates non-normal residuals, and thus a failure of ANOVA assumptions.

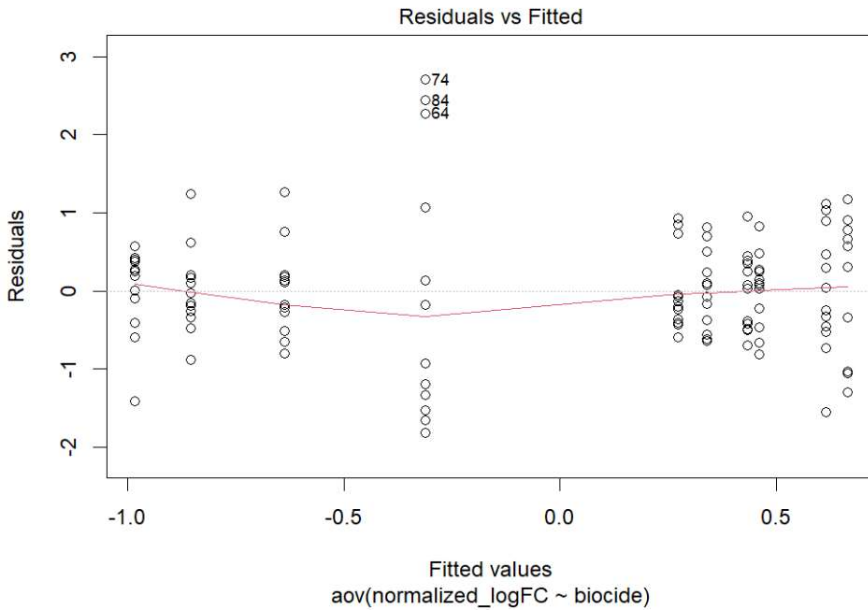


Figure S5. Residuals vs Fitted plot for the ANOVA assumptions testing of the “normalised logFC values against gene family” linear model. While the non-linear slope and unbalanced spread of residuals indicates inhomogeneous variance (and thus a failure of ANOVA assumptions) this is a major improvement over the Residuals vs Fitted plot utilising the non-normalised logFC values (S3).

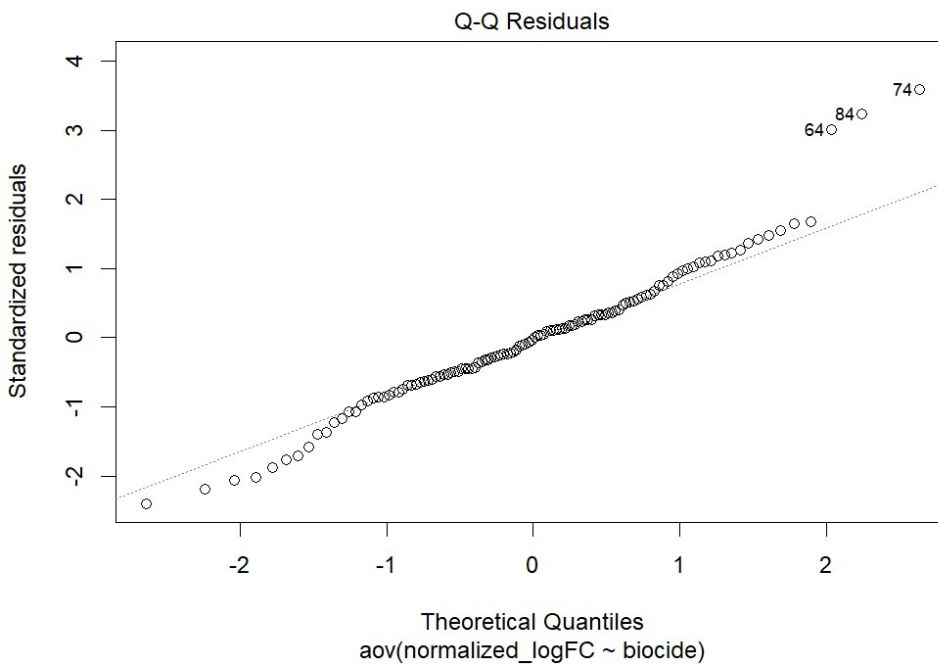


Figure S6. Q-Q plot for the ANOVA assumptions testing of the “normalised logFC values against gene family” linear model. The relative linearity of the plot indicates normal residuals, and thus a tentative fulfillment of the ANOVA assumptions. This is a major improvement over the Normal Q-Q plot utilising the non-normalised logFC values (S4).

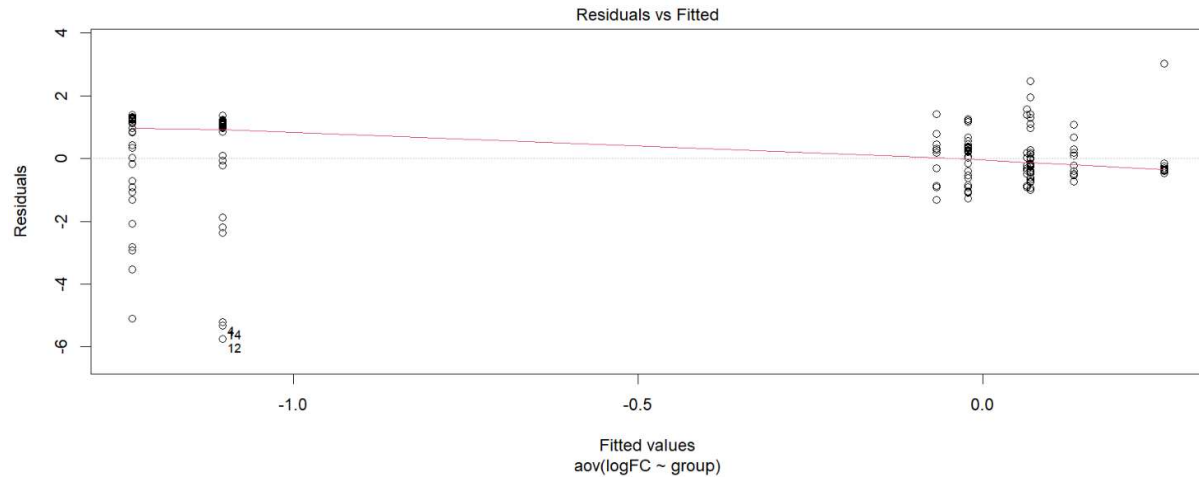


Figure S7 Residuals vs Fitted plot for the ANOVA assumptions testing of the “logFC values against gene group” linear model. This model is similar to that referenced in **S3** and **S4**, however now the gene groups factor contains the associative genes *zur* and *mlcA*, as well as the hypothetical proteins *BAL062_00181* and *BAL062_1982*. The unbalanced spread of residuals and negative slope of the line indicates inhomogeneous variance, and thus a failure of ANOVA assumptions.

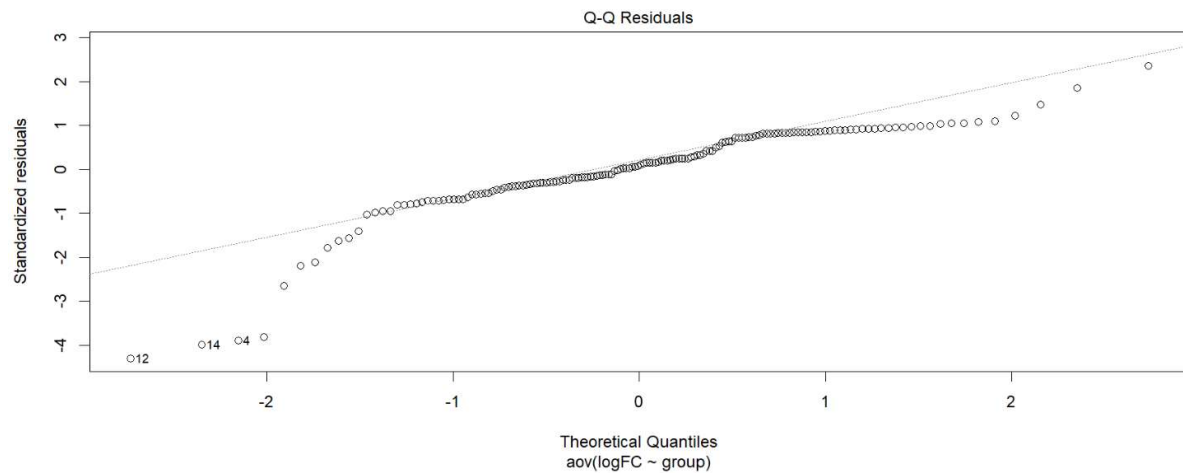


Figure S8. Q-Q plot for the ANOVA assumptions testing of the “logFC values against gene group” linear model. This model is similar to that referenced in **S3** and **S4**, however now the gene groups factor contains the associative genes *zur* and *mlcA*, as well as the hypothetical proteins *BAL062_00181* and *BAL062_1982*. The lack of linearity indicates non-normal residuals, and thus a failure of ANOVA assumptions.